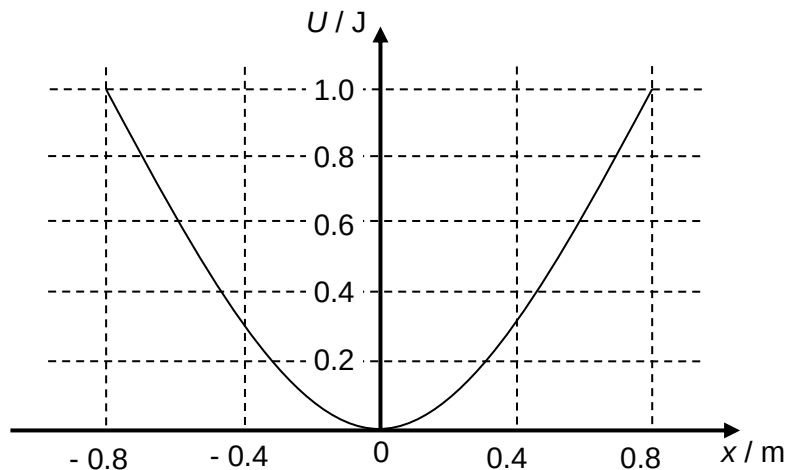


- 10 A particle of mass 5.0 kg moves with simple harmonic motion and the variation with position x of its potential energy U is shown below.



What is the period of oscillation of the mass?

A 7.9 s

B 8.8 s

C 11 s

D 20 s

10	A	$KE_{\max} = U_{\max} = 1.0$ $\frac{1}{2}mv_{\max}^2 = 1.0$ $v_{\max} = 0.632$ $v_{\max} = \omega x_0 = \frac{2\pi}{T} x_0$ $T = \frac{2\pi \times 0.8}{0.632} = 7.9 \text{ s}$
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- 11 A double star system is at a distance of 1.0×10^{17} m from the Earth. A telescope with a